# Case

Patient demographics

Musculoskeletal Physiotherapy

## ■ SUBJECTIVE PROBLEM LIST

- Physical Impairments
- SIJ pain right-sided (VAS 6/10)
- Reduced lumbopelvic motor control
- Possible pelvic floor weakness
- Unable to perform single-leg yoga postures
- Difficulty rolling in bed / disrupted sleep
- Pain exiting a car (asymmetric load)
- Unable to teach / demonstrate yoga fully
- Heightens pain sensitivity (central sensitisation risk) and reduces capacity for exercise adherence.
- Ask "How are you feeling emotionally overall?" — PND co-presents with chronic pain and impairs recovery.

## ■■ RED FLAGS

- Night pain / rest pain not relieved by position change:
- May indicate sinister pathology (infection, malignancy). Action: urgent medical referral.
- Bilateral leg weakness or saddle anaesthesia:
- Cauda equina syndrome — a surgical emergency. Action: immediate emergency referral; do not proceed with assessment.
- Unexplained significant weight loss (>5kg over weeks):
- Possible malignancy or systemic disease. Action: urgent medical referral.
- Fever / systemically unwell:
- Suggests infection (e.g. septic arthritis, discitis — postpartum women are at elevated risk). Action: urgent medical referral.

## ■ PATHOPHYSIOLOGY (KEY POINTS)

- Working Diagnosis: Postpartum SIJ Dysfunction / Pelvic Girdle Pain During pregnancy, the hormone relaxin is secreted to increase ligamentous laxity across the pelvis, facilitating delivery.
- Postpartum, circulating relaxin decreases but remains elevated during breastfeeding; concurrently, the postural and neuromuscular demands of pregnancy alter motor recruitment patterns, leaving the force-closure system temporarily deficient.
- The result is excessive translational and rotational shear at the SIJ, particularly during unilateral loading.
- The posterior SIJ ligaments and the posterior interosseous ligament become the primary stabilisers under load — their repeated stress-straining generates localised nociception, perceived as deep buttock/PSIS pain.
- Activities that increase torsional forces (rolling in bed, stepping out of a car, single-leg stance) dramatically increase SIJ shear forces, explaining the classic provocation pattern.
- Unilateral PSIS tenderness: Posterior SIJ ligament and joint capsule irritation from excessive shear.

## ■ DIFFERENTIAL DIAGNOSES

See full case notes

## ■ OBJECTIVE ASSESSMENTS & HOW TO PERFORM

**Key Observation Findings**

Observe gait: antalgic gait, Trendelenburg sign (gluteus medius weakness), reduced step length on right Standing posture: posterior pelvic tilt, asymmetric PSIS height (palpate bilaterally), increased lumbar lordosis Active movement: observe for lateral trunk shift during single-leg stance, hip hitching pattern

**Thigh Thrust Test (Posterior Shear Test)**

What it measures: Posterior SIJ ligament integrity and joint stability under posterior shear force. High sensitivity (88%) for SIJ pain. How to perform: Patient lies supine on the plinth. Flex the test hip to 90° and slightly adduct (approximately 10°).

**Sacral Thrust Test**

What it measures: Anterior shear stress across both SIJs simultaneously — tests the integrity of the anterior SIJ ligaments and evaluates bilateral posterior ligament stress. How to perform: Patient lies prone on the plinth, arms by sides. Stand to the side. Place both hands stacked over the sacrum (at S2 level — midway between the posterior superior and posterior inferior iliac spines).

**Active Straight Leg Raise (ASLR)**

What it measures: Force-closure capacity of the lumbopelvic region; specifically assesses whether the patient can transfer load through the pelvis efficiently. A key functional test in postpartum PGP research (Mens et al.). How to perform: Patient lies supine, legs straight, arms by sides. Ask the patient to lift one leg (keeping it straight) approximately 20cm off the plinth without bending the knee.

## ■ TREATMENT PLAN

- Pelvic Girdle Questionnaire (PGQ)
- Exercise 1: Transversus Abdominis / Deep Core Activation ("Drawing In" with Pelvic Floor Co-contraction)
- Reduce hold time to 5 seconds; use ultrasound biofeedback if difficulty with isolation; perform in supported sitting if crook-lying is uncomfortable.
- Add gentle heel slide while maintaining activation → progress to 4-point kneeling with arm/leg reach → progress to sitting/standing functional contexts.
- Exercise 2: Clamshell (Hip External Rotation, Gluteus Medius Activation)
- Reduce range of motion; perform against a wall to cue neutral pelvis; reduce sets to 2×10.
- Add resistance band above knees → side-lying hip abduction (straight leg) → standing hip abduction with band → single-leg mini-squat with pelvic control cue.

• Technique: Sacroiliac Joint Mobilisation (Grade I–II Posterior-Anterior Glide) + Soft Tissue Release to Posterior SIJ Ligaments

• Soft tissue release (sustained pressure × 60–90 seconds) to the posterior SIJ ligament and gluteal trigger points can reduce guarding and facilitate subsequent exercise.

• Patient Education — 4 Key Messages

• "Your pelvis is currently hypermobile, not damaged."

• The ligaments softened during pregnancy to allow birth — they are healing, not torn. This reframes pain as a temporary instability rather than injury, reducing fear-avoidance.